Online Banking System with Virtual Wallet Management

API & Microservices – Review 1

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24B11AI373

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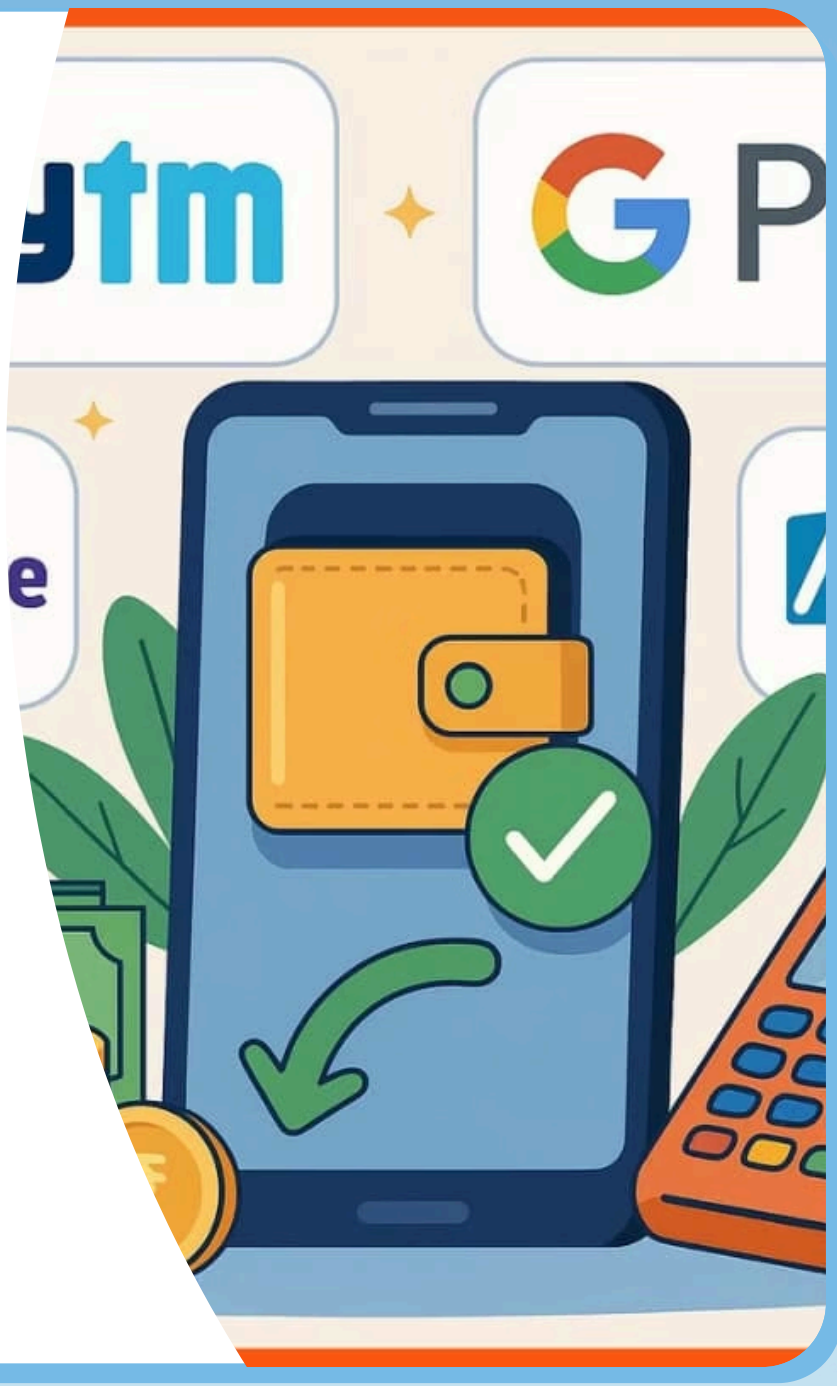
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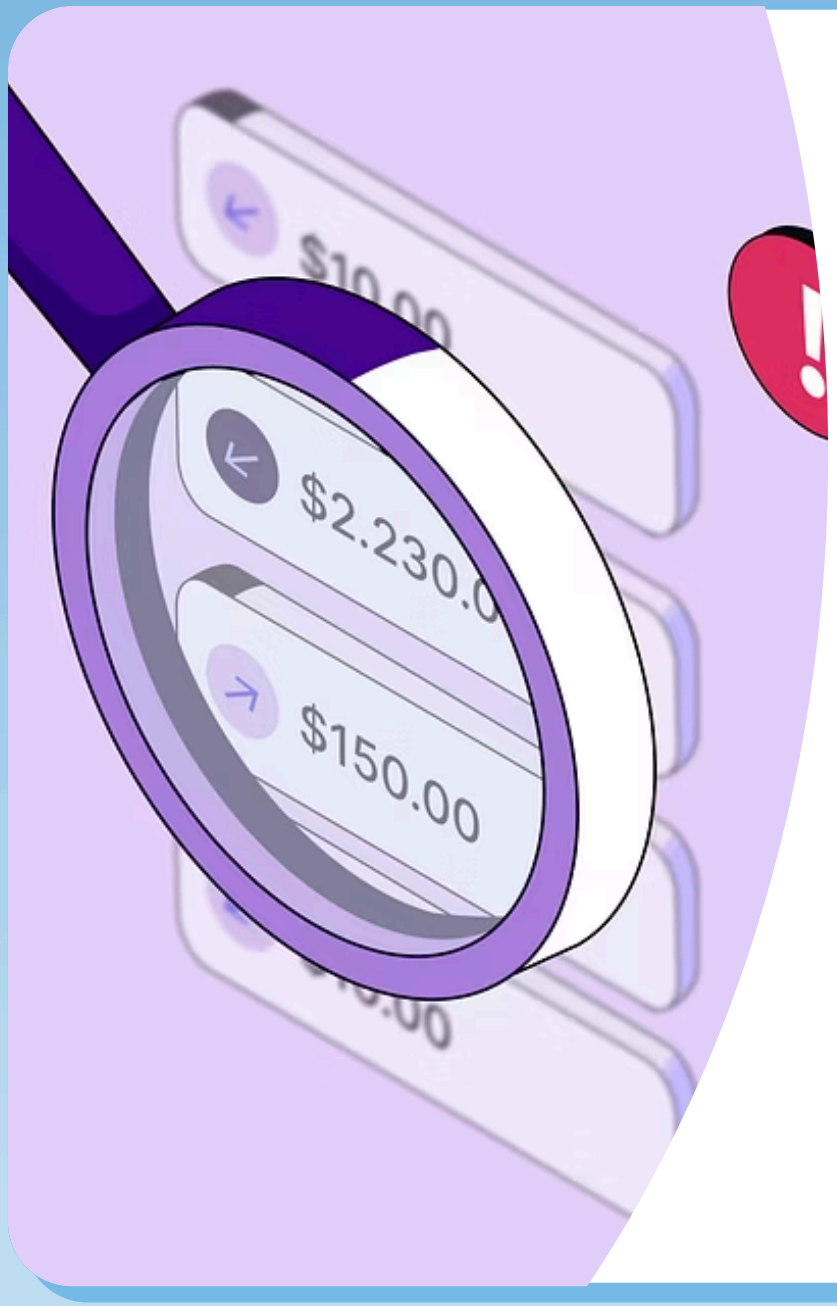
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Problem Identification

Conventional banking shows a **single available balance** — but users mentally divide money across categories like Savings, Food, Travel, and Shopping. Standard records cannot track **wallet-wise spending behaviour**.

What We Want to Measure

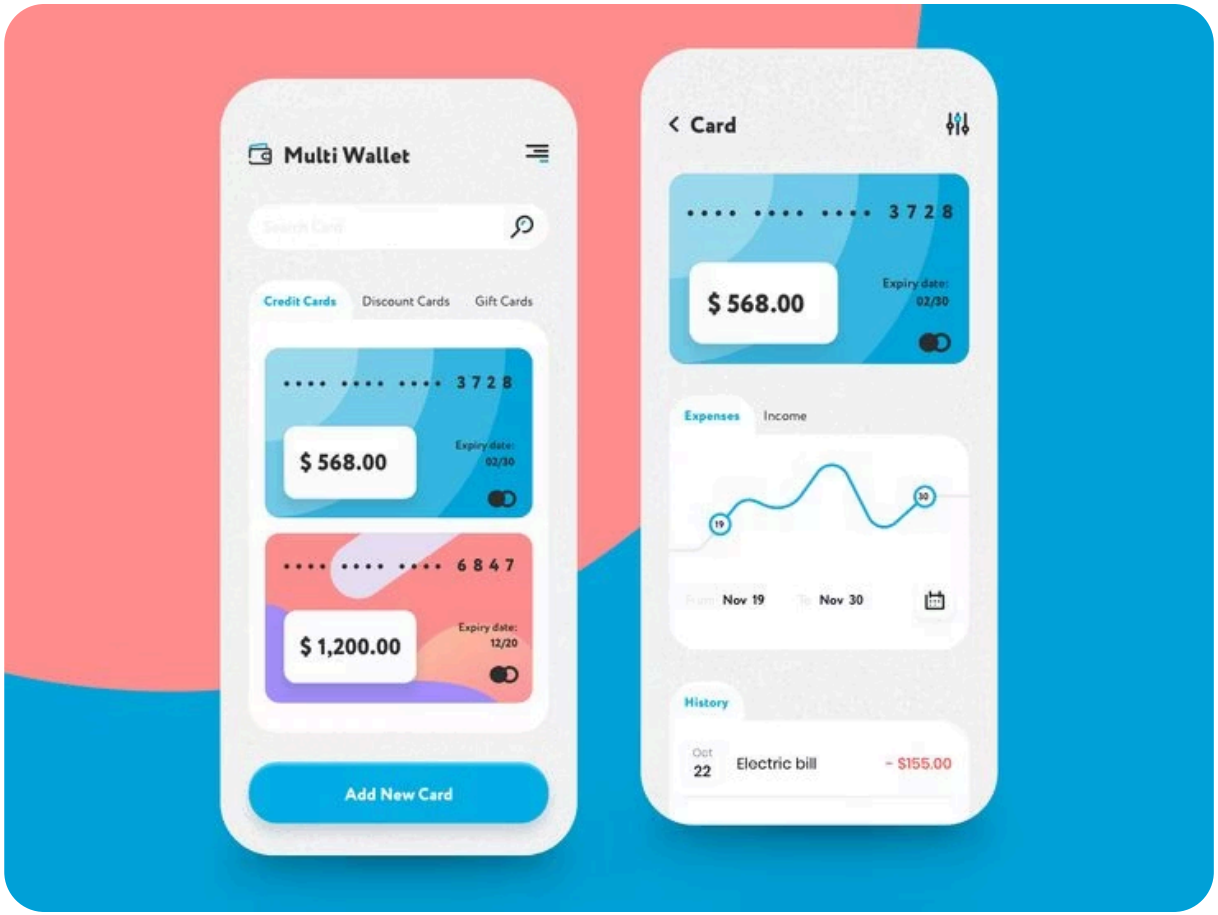
- Average transaction amount per wallet
- Spending variation across wallets
- Distribution of transaction amounts
- Probability a transaction exceeds a threshold

The Core Problem

A virtual-wallet banking system can serve as a **real-world data source for Probability & Statistics** — turning everyday transactions into meaningful statistical observations.

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Proposed Solution: Virtual Wallet-Based Banking



How It Works

Users divide their total balance into **logical virtual wallets** — not separate accounts. The sum of all wallets always equals the total available balance.

If a wallet is insufficient, the **General Wallet** covers the remainder. Every transaction is logged with wallet, amount, date, and description — forming the P&S dataset.

Savings ₹3,000

Food ₹2,000

Travel ₹1,500

Shopping ₹1,500

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General ₹2,000

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Why API Instead of a Normal Application?

A traditional app couples frontend, logic, and database. Our system **decouples the client** from banking and statistical operations via REST API.



Separation

UI contains no banking logic or statistical calculations.



Reusability

Same API serves web, mobile, or third-party clients.



Security

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Clients access controlled endpoints, not the database directly.

Maintainability

Wallet, transaction, and stats modules evolve independently.

Where Probability & Statistics Comes In

Every payment is a **statistical observation**. Transaction amounts form sample data that can be rigorously analyzed.

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Descriptive Statistics

- Mean, Median, Variance
- Standard Deviation, Range
- Quartiles & Percentiles

Probability Questions

- P(transaction > ₹500)?
- P(amount in a spending range)?
- Spending distribution per wallet

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Statistical Analysis of Wallet Data

1

Mean

Average transaction amount.
e.g., Food: 200, 300, 400, 100 → Mean = **₹250**

2

Median

Middle value after sorting. Robust against outliers that skew the average.

3

Variance

Measures how far transaction amounts spread from the mean.

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4

Standard Deviation

Low SD → consistent spending. High SD → considerable variation in amounts.

Probability Distribution of Transactions

Random Variable X = Transaction Amount

Suppose the Food Wallet has 19 transactions distributed across ranges:

Range	Transactions
₹100 – ₹200	5
₹200 – ₹300	8
₹300 – ₹400	4
₹400 – ₹500	2

P(₹200 ≤ X < ₹300) = 8/19 ≈ 0.421
~42.1% of transactions fall between ₹200 and

What the System Provides

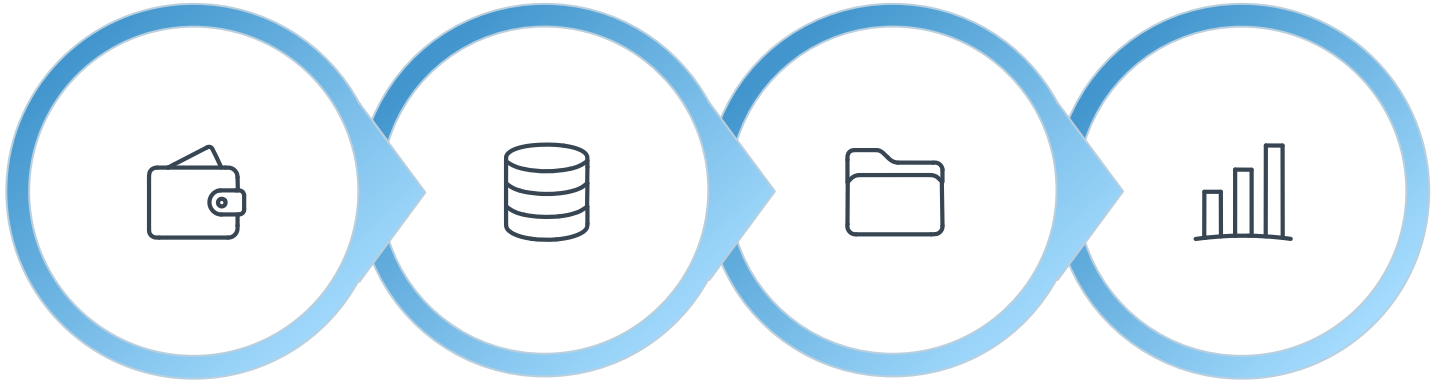
- **Transaction probability by range**
- **Spending distribution per wallet**
- **Frequency of transaction amounts**
- **Probability of exceeding a threshold**

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₹300.

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Complete P&S Workflow



User Payment

Process & Store

Group by Wallet

Generate P&S Metrics

The system transforms every user payment into a structured statistical dataset, grouped by wallet, and outputs metrics such as transaction count, mean, median, variance, standard deviation, and probability results — all accessible via the API.

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Module 1: User & Account Management

Purpose: Manages the user's identity and the main banking account from which virtual wallet allocations are maintained.

Main Functions:



User registration and login



User authentication



Account creation and management



Maintain total available balance



Display account information

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Maintain relationship between user and wallets

Example:

User: Karthik

Account Balance = ₹10,000

↓

Virtual Wallet Allocation

Savings: ₹3,000

Food: ₹2,000

Travel: ₹1,500

General: ₹3,500

Role in System: This module provides the base account and total balance from which the virtual wallet system operates.



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Module 2: Virtual Wallet Management

Purpose: Allows users to divide and manage their available money through logical virtual wallets.

Main Functions:

- Create virtual wallets
- Set wallet names/categories
- Allocate money to wallets
- Modify wallet allocation
- Transfer allocation between wallets
- Maintain wallet balances

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Provide a default General Wallet

Example:

Total Balance = ₹10,000

Savings ₹3,000

Food ₹2,000

Travel ₹1,500

Shopping ₹1,000

General ₹2,500

The system ensures: Σ Wallet Balances = Total Available Balance

Role in System: The wallet module provides the main differentiating feature of the project: users can control how their available money is logically distributed.



Module 3: Transaction Management

Purpose: Handles payments and records which virtual wallet is used for each transaction.

Main Functions:

- Record transactions
- Select payment wallet
- Check wallet balance
- Deduct transaction amount
- Handle insufficient wallet balance
- Use General Wallet for overflow
- Ask user to select another wallet when required
- Maintain wallet-wise transaction history

Example:

- Food Wallet = ₹2,000
- Payment = ₹3,000

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- Food Wallet → ₹2,000 (fully used)
- Remaining → ₹1,000
- General Wallet (covers ₹1,000)
- If General Wallet is insufficient, user selects another wallet.

Role in System: This module generates the transaction dataset that is later used by the Probability & Statistics module.

Module 4: Probability & Statistics Analytics

Purpose: Uses the transaction data generated by the banking system to provide statistical metrics for each wallet.

Input Data:

- Wallet
- Amount
- Date

Metrics Calculated:

- Number of transactions
- Total spending
- Mean

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- Transaction Type

- Median
- Variance
- Standard deviation
- Transaction frequency/distribution
- Relevant probability measures

Example Output - Food Wallet:

Transactions:
25

Total Spent:
₹6,250

Mean:
₹250

Median:
₹220

Variance:
[calculated]

Standard Deviation:
[calculated]

P(X > ₹500):
[calculated]

Role in System: This is the Probability & Statistics component of the project. The system calculates and presents statistical measures based on collected transaction data.

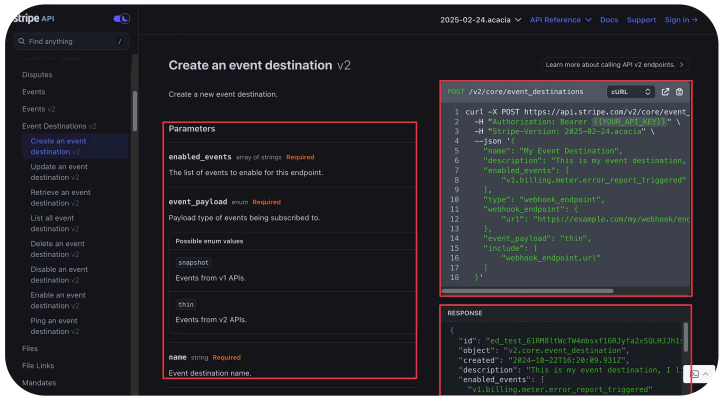
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Module 5: API & Service Management

Purpose: Provides a common communication layer between the client application, banking modules, database and statistical processing.

Major API Responsibilities:

1. **Client Request**
2. **API Validation**
3. **Business Module**
4. **Database**
5. **API Response**



Example APIs:

- POST /api/auth/login
- POST /api/wallets
- POST /api/transactions
- GET /api/wallets
- GET /api/analytics/{wallet}

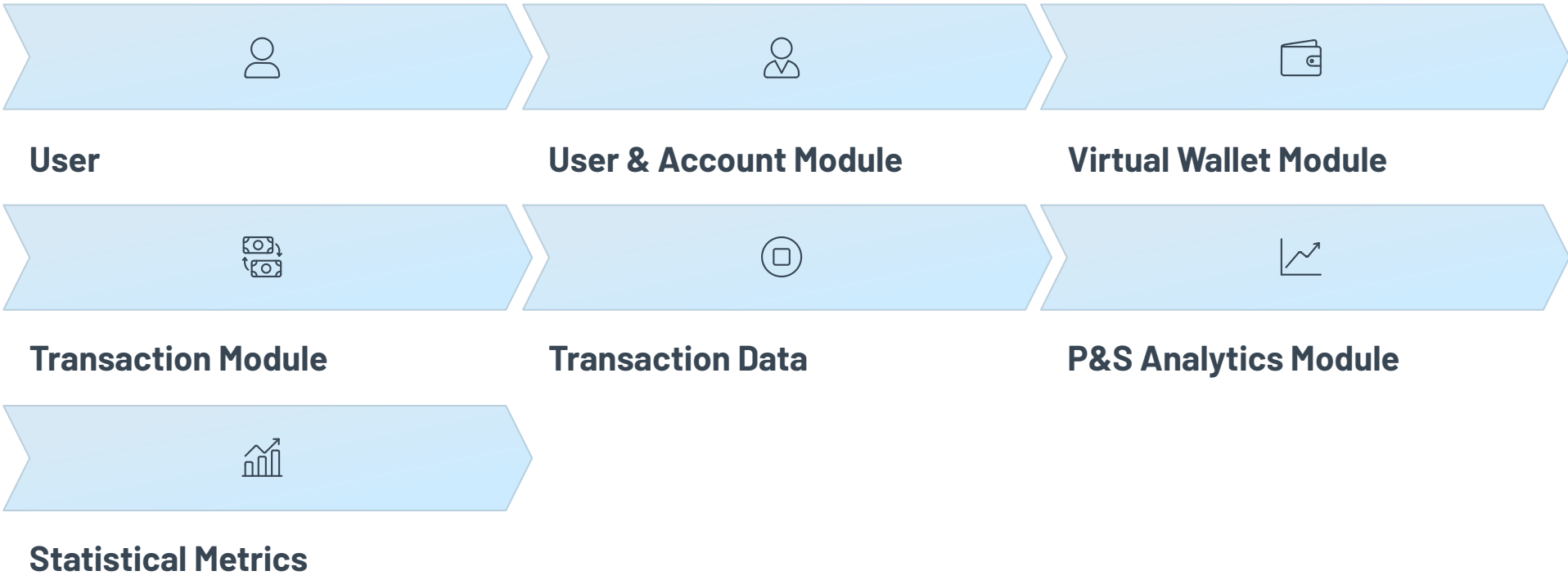
Architecture: Frontend → API → Module → Database

Role in System: The API provides a standard, secure and reusable interface to the entire system instead of the frontend directly accessing each module or the database.

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How the Major Modules Work Together

Module Flow:



API Integration:

- REST API connects all modules:

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- Account Module
- Wallet Module
- Transaction Module
- P&S Module
- Database

Overall Logic:

Account provides the money → Wallets organize it → Transactions consume it → Transaction data feeds P&S → P&S provides statistical metrics → API connects and exposes all these functions to the client.

This architecture gives a clean module structure: 5 major modules, each with a clear purpose, without cluttering the presentation with minor components.

End-to-End API Flow

Recording a ₹300 Food Transaction

01

Client Sends Request

`POST /api/transactions` with wallet, amount, description

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02

API Validates & Processes

Checks user, wallet balance, then processes payment

03

Database Updated

Food Wallet: ₹2,000 → ₹1,700; transaction logged

04

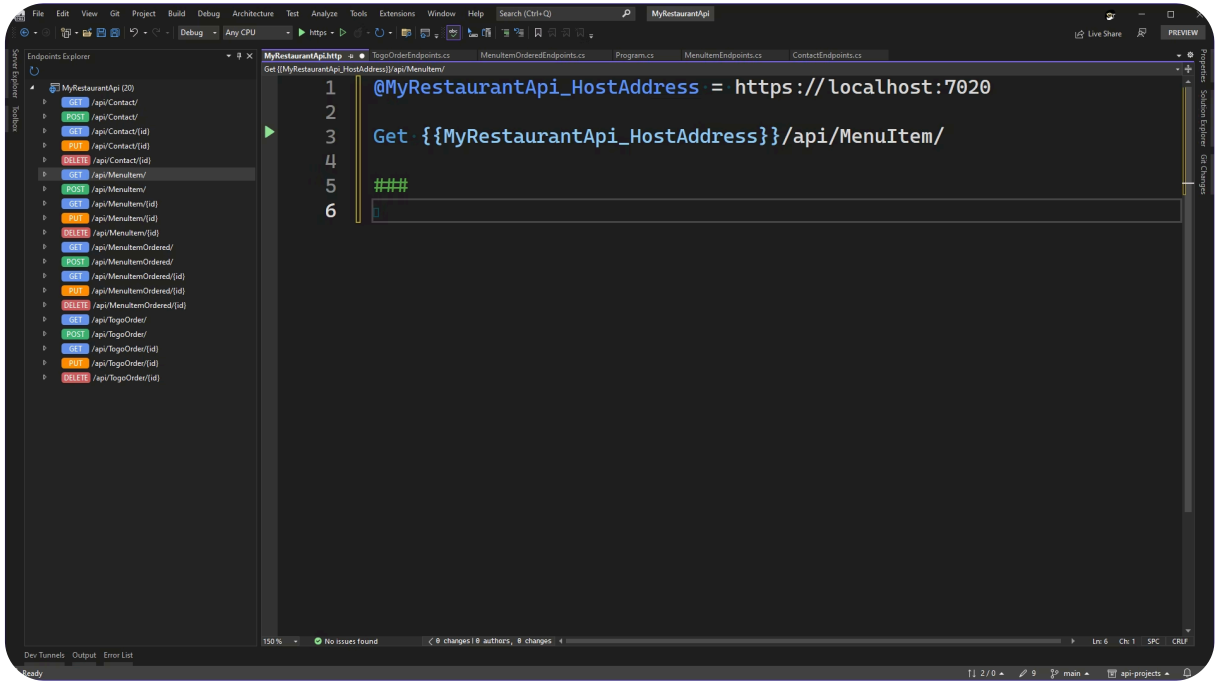
Response Returned

JSON with status, wallet, amount, remaining balance

05

Stats API Called

`GET /api/analytics/wallet/food` returns mean, median, variance, SD, probability



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